Can 100 Real Interviews Rescue a Synthetic Subgroup Estimate?

100 real interviews + approach C · a short simulation

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The short version

We simulated a cheap, quota-free survey by drawing 50 to 400 people from CES 2025, then combined their answers with approach C, the dossier model that directly predicts distributions for 15 region-by-age groups (C trained by example, first training run; the best C so far). The full CES 2025 sample is the answer key. Each condition is repeated 100 times; the score is total variation (TV), roughly the share of answers put in the wrong category. Lower is better.

The hybrid works. With **100 biased real interviews**, the best version scores **0.106 on repeated questions and 0.115 on new questions**, against 0.315 and 0.322 for the small sample alone. That is about as accurate as a **balanced, representative survey of 500–650 people** (about ±16–17 points per group). With **400 biased interviews**, it reaches **0.081 and 0.089**, about a representative survey of **830–1,060 people**, the size of a standard national poll (about ±12–13 points per group).

Correction (October 20). A first version reported much higher equivalences ("1,200 interviews"). Several CES 2025 questions were asked of only 25–50% of respondents, so a simulated "400-person" collection held only 100–200 answers on those questions, which made the sample alone look far worse than a real survey where everyone answers. Each question's sample is now drawn among the respondents who were asked it. All numbers below use the corrected simulation.

What was compared

- **Sample only:** weighted answers from the small real sample, post-stratified to the population shares of the 15 groups. A group with no answers receives the sample's national distribution.
- **C only:** the mean of three existing approach-C draws; the model was not called again.
- **Bayesian hybrid:** C is a Dirichlet prior and the real answers update it. Prior strength was selected on the repeated questions; **k = 20** won for every sample size and sampling mechanism.
- **National level + C gaps:** the real sample, post-stratified by region and age, supplies the national answer distribution; C supplies the differences among groups.

The biased collection deliberately gives twice the selection chance to respondents reporting high political interest (8–10 on a 0–10 scale). It uses no age or region quotas and does not correct for political interest afterward. This is a mild bias; a real quota-free collection can be much worse.

Results

Repeated questions

Random collection

Method	n=50	n=100	n=200	n=400
Sample only	0.387	0.314	0.235	0.167
C only	0.151	0.151	0.151	0.151

Method	n=50	n=100	n=200	n=400
Bayesian hybrid (k=20)	0.142	0.135	0.124	0.108
National level + C gaps	**0.132**	**0.104**	**0.088**	**0.079**

Biased collection

Method	n=50	n=100	n=200	n=400
Sample only	0.382	0.315	0.238	0.169
C only	0.151	0.151	0.151	0.151
Bayesian hybrid (k=20)	0.142	0.134	0.123	0.107
National level + C gaps	**0.132**	**0.106**	**0.090**	**0.081**

New questions, including the two Trump questions

Random collection

Method	n=50	n=100	n=200	n=400
Sample only	0.389	0.322	0.243	0.171
C only	0.213	0.213	0.213	0.213
Bayesian hybrid (k=20)	0.197	0.185	0.165	0.140
National level + C gaps	**0.138**	**0.113**	**0.096**	**0.087**

Biased collection

Method	n=50	n=100	n=200	n=400
Sample only	0.388	0.322	0.247	0.174
C only	0.213	0.213	0.213	0.213
Bayesian hybrid (k=20)	0.197	0.185	0.167	0.141
National level + C gaps	**0.141**	**0.115**	**0.098**	**0.089**

What it is worth, in survey terms

Equivalent size = the balanced, representative survey (everyone answers every question) with the same TV, read on the simulated sample-only curve for balanced surveys; the conversion of the October 14 note (TV ≈ 0.66 / √n per group) gives about 10–15% less. Margin of error per group at 95%, for a proportion near 50%.

Biased collection + C	TV, repeated / new	Worth a representative survey of	Margin per group
100 interviews	0.106 / 0.115	500–650	±16–17 pts
400 interviews	0.081 / 0.089	830–1,060	±12–13 pts
For reference: standard poll of 1,000, balanced	*≈ 0.08*	*1,000*	*±12 pts*

The Trump-question sensitivity check

The two questions about Canada–US relations after Trump's election are exactly where a pre-2025 model cannot know the current national level. At n=100 in the biased collection:

Method	All 8 new questions	Other 6 new questions
Sample only	0.322	0.373
C only	0.213	0.134
Bayesian hybrid (k=20)	0.185	0.131
National level + C gaps	**0.115**	**0.127**

Without the Trump questions, C alone is already strong and 100 interviews add little. With them, real data supply the missing 2025 level and the hybrid improves sharply: the small sample matters most when opinion has moved since the model's data.

Four conclusions

1. **The hybrid beats the sample alone at every size tested**, from n=50, for repeated and new questions.
2. **100 biased interviews + C are worth a representative survey of about 500–650 people; 400 + C, about 830–1,060**, the size of a standard national poll.
3. **This mild selection bias barely survives the mix** (at most 0.003 TV against random collection); a stronger or topic-related bias remains untested.
4. **The level-plus-gaps hybrid levels off near 0.07–0.08** as the sample grows: that floor is the error of C's own group differences. Beyond about 800 interviews the Bayesian hybrid, which lets real data override C cell by cell, becomes better (0.06–0.07 at n=1,600). Better group differences from C are what would lower the floor.

The simulated sample overlaps the CES 2025 answer key by design, so the largest-sample results are slightly optimistic. The optional Quebec age-by-gender variant was not run because the existing C outputs distinguish region and age, but not gender. Full repetition-level results are in `data/analysis/hybrid_small_sample.csv`; the reproducible analysis is `scripts/44_hybrid_small_sample.py`.